

```
# Perceptron Based Student Classification
```

```
from sklearn.linear_model import Perceptron
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Dataset
```

```
# [Study Hours, Attendance]
```

```
X = [
```

```
    [2, 60], [3, 65], [4, 70], [5, 75],
```

```
    [6, 80], [7, 85], [8, 90], [9, 95],
```

```
    [1, 50], [2, 55], [3, 60], [4, 65],
```

```
    [6, 75], [7, 80], [8, 85], [9, 90]
```

```
]
```

```
# 0 = Fail, 1 = Pass
```

```
y = [
```

```
    0, 0, 0, 0,
```

```
    1, 1, 1, 1,
```

```
    0, 0, 0, 0,
```

```
    1, 1, 1, 1
```

```
]
```

```
# Split data
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y,
```

```
    test_size=0.25,
```

```
    random_state=42
```

```
)
```

```
# Create Perceptron model

model = Perceptron(max_iter=1000, random_state=42)

# Train model

model.fit(X_train, y_train)

# Predict

y_pred = model.predict(X_test)

# Accuracy

accuracy = accuracy_score(y_test, y_pred)

print("Student Pass/Fail Classification")
print("-----")
print("Accuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)

# Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Display confusion matrix

plt.figure(figsize=(5, 4))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
```

```
cmap="Blues",  
xticklabels=["Fail", "Pass"],  
yticklabels=["Fail", "Pass"]  
)  
  
plt.xlabel("Predicted")  
plt.ylabel("Actual")  
plt.title("Perceptron - Student Classification")  
plt.show()
```

```
Student Pass/Fail Classification  
-----  
Accuracy: 0.5  
Accuracy (%): 50.0  
  
Confusion Matrix:  
[[0 2]  
 [0 2]]
```

